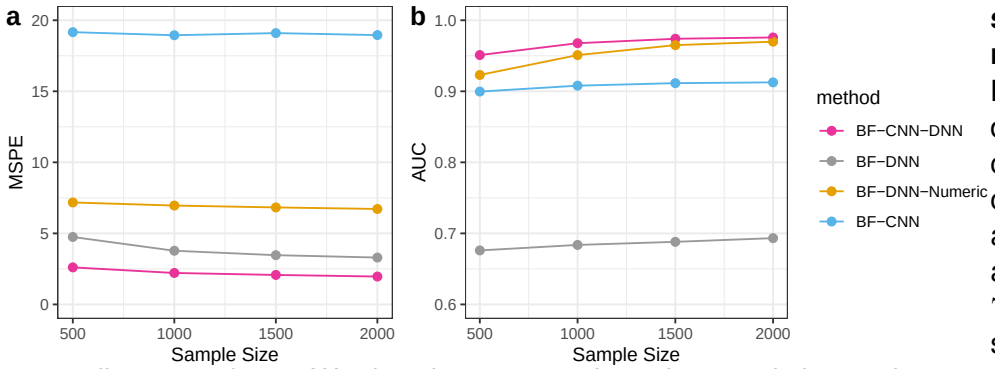
$\mathrm{Bin}(1, 0.5)$. We simulate the outcome in two ways: a) continuous $Y_i = f(s_i, d_i, h_i, W_i) + \epsilon_i$, $\epsilon_i \sim \mathrm{N}(0, 1)$; and b) binary $Y_i = \mathbf{I}(\mathrm{f(s_i, d_i, h_i, W_i)} + \varepsilon_\mathrm{i} > 7)$, where $f(s_i, d_i, h_i, W_i) = s_i(1+W_{i1}+W_{i6})/2+10|d_i-0.15|+10h_i+W_{i2}^2$ and $\varepsilon_i \sim \mathrm{N}(0, 1)$. We compare the proposed BF-CNN-DNN framework with three other deep learning-based methods: 1) BF-DNN, where the 7×24 POP profile image is transformed to a vector with 168 elements, along with all other clinical variables, as the input to a BF-DNN model; 2) BF-DNN-Numeric, where only the clinical variables are used as the input to the BF-DNN model; 3) BF-CNN, where only the 7×24 POP profile images are used to fit the BF-CNN model. BF-CNN-DNN *distinctly outperforms the other three methods* as shown in Figure 5.

Figure 5: Preliminary Results of Aim 1b



Model validation, expected results, potential pitfalls and alternative plans: We will apply the BF-CNN-DNN method to TEMPOS data and predict the trajectory of chronic POP. Specifically, we will convert each patient's first week acute POP profile into hourly averages and then encode them into a 7×24 image with each row representing a day and each column corresponding to an hour. We then integrate patient characteristics such as age, gender, surgery duration, anesthesia method, body mass index (BMI), etc, with the converted acute POP profile image into the BF-CNN-DNN framework. We expect to *remarkably improve chronic POP prediction accuracy*.

Model validation will be conducted via 10-fold cross-validation. To investigate the generalizability of the proposed model, the R^2 or area under receiver-operating curve (AUC) for each test fold will be generated and the mean R^2 or AUC by combining the results of each fold will be computed. For the TEMPOS data with relatively small sample size, the BF-CNN-DNN methods may have data deficits that limit their performance. If this happens, we will adopt several strategies, including transfer learning and domain adaptation strategies [118, 119] using the modeling results from the CDW-H data, e.g. using large CDW-H acute POP data to train the BF-CNN-DNN model.

Aim 2: Develop Efficient Deep Learning Methods to Investigate the Causal Mechanisms of POP

Overview and rationale: Confounding widely exists in observational data (e.g., EHR data) and the confounding structures in POP EHR data can very complicated including the non-linear interactions among confounding variables [14, 40]. The confounding variables can be the initial pain level, acute pain trajectory, demographic factors, opioid use, operative features, and many more. The work of Zheng et al. [45] demonstrated that the gender effects on POP outcomes (e.g., severe POP duration and postoperative opioid consumption) are confounded by age and preoperative pain factors. The commonly used methods to adjust for confounding factors are propensity score (PS) related methods [120]. Another strategy is to employ the kernel regression based semiparametric framework of Robinson [67]. These methods either *heavily* rely on parametric modeling assumptions or are *sensitive* to the curse of dimensionality [121]. Existing mediation analysis methods [122–128] share similar concerns. To address these analytical challenges by adjusting for the complex confounding structures, we will develop a set of BF-DNN-based semi-parametric models as described below by taking the advantages of BF-DNN methods for *universally* approximating complex functions accurately [60, 61].

Aim 2a: BF-DNN semi-parametric comparative effectiveness analysis framework Let Y be an outcome (e.g., POP intensity), X be a set of covariates (e.g., age, gender, BMI, surgery type and duration), and Z be the factor of primary interest (e.g., anesthesia method). To deal with the complex confounding, the following semi-parametric model: $Y = Z\tau(\mathbf{X}) + \gamma(\mathbf{X}) + \epsilon$ has been proposed and investigated [67, 129], where τ and γ are (unspecified) functions of covariate vector $\mathbf{X}$. In this model, $\tau(\mathbf{X})$ refers to the treatment effect of Z. The models with constant treatment effect, or $\tau(\mathbf{X}) = \beta$ have been investigated by Robinson [67], while Nie and Wager [129] investigated the case where the form of $\tau(\mathbf{X})$ is unspecified, which allows unspecified interactions between causal factors and confounding covariates [129]. The random error $\epsilon \perp (Z, \mathbf{X})$, with $E(\epsilon) = 0$ and $Var(\epsilon) = \sigma^2$.

For cases with $\tau(\mathbf{X}) = \beta$, Robinson [67] proposed an alternative semi-parametric model $Y - E(Y|\mathbf{X}) = \beta(Z - E(Z|\mathbf{X})) + \epsilon$ for estimating β where $e(\mathbf{X}) \equiv E(Z|\mathbf{X})$ and $\xi(\mathbf{X}) \equiv E(Y|\mathbf{X})$ are estimated by their Nadaraya-Watson kernel estimators. The semi-parametric regression estimator of β enjoys a root-n-consistency [67]. However, it relies on the accuracies of the estimates of $\xi(\mathbf{X})$ and $e(\mathbf{X})$. Kernel-based methods can perform poorly under high-

dimensional settings due to *the curse of dimensionality* [121] and involve *very intensive computational workloads*. To overcome these issues, we adopt BF-DNN to robustly estimate $\xi(\mathbf{X})$ and $e(\mathbf{X})$. However, when β is not small, the error distribution associated with $Y|\mathbf{X}$ becomes bimodal which can heavily impact the estimate of $\xi(\mathbf{X})$. We propose a revised semi-parametric procedure where $Y^* - \xi^*(\mathbf{X}, \beta_1) = (\beta - \beta_1)(Z - E(Z|\mathbf{X})) + \epsilon$ is modeled. Here, we have $Y^* = Y - \beta_1 Z$ for a given constant β_1 and $\xi^*(\mathbf{X}, \beta_1) = E(Y^* \mid \mathbf{X})$. This Double-BF-DNN procedure (summarized in Algorithm 1) leads to the following treatment effect estimate and associated variance estimator: $\widehat{\beta}^* = \frac{\sum_{i=1}^n \{Y_i - \beta_1 Z_i - \widehat{\xi}^*(\mathbf{X}_i, \beta_1)\}\{Z_i - \widehat{e}(\mathbf{X}_i)\}}{\sum_{i=1}^n \{Z_i - \widehat{e}(\mathbf{X}_i)\}^2} + \beta_1$ and $\widehat{V}(\widehat{\beta}^*) = \frac{\widehat{\sigma^2}^*}{\sum_{i=1}^n \{Z_i - \widehat{e}(\mathbf{X}_i)\}^2}$, where $\widehat{\sigma^2}^* = \sum_{i=1}^n [Y_i - \beta_1 Z - \widehat{\xi}^*(\mathbf{X}_i, \beta_1) - \widehat{\beta}^*\{Z_i - \widehat{e}(\mathbf{X}_i)\}]^2/n$. If β_1 is selected close to β, the revised procedure can substantially improve the treatment effect estimate. In our preliminary study, we set β_1 to the estimated β derived from the covariate adjustment by PS which works reasonably well, but other options will be investigated. For example, we may try a set of β_1s and pick the optimal one that minimizes $\widehat{V}(\widehat{\beta})$. For cases where $\tau(\mathbf{X})$ is unspecified, often the average treatment effect (ATE) of Z, or $\tau^* = E_{\mathbf{X}}(\tau(\mathbf{X}))$ is of primary interest. Several heterogeneous treatment effects learning algorithms have been proposed to estimate $\tau^* = E_{\mathbf{X}}(\tau(\mathbf{X}))$ [129]. We develop the following Triple-BF-DNN based counterpart of the algorithms in Nie and Wager [129] as follows: $Y - \xi(\mathbf{X}) = (Z - e(\mathbf{X}))\tau(\mathbf{X}) + \epsilon$ with $\xi(\mathbf{X}) \equiv E(Y \mid \mathbf{X})$ and $e(X) \equiv E(Z \mid \mathbf{X})$. To estimate $\tau(\mathbf{X})$, we first apply BF-DNN to get $\hat{\xi}(\mathbf{X})$ and $\hat{e}(\mathbf{X})$, i.e., estimates of $\xi(\mathbf{X})$ and $e(\mathbf{X})$, respectively. After plugging these estimates into the semi-parametric model above, we apply BF-DNN a third time to get the estimate of $\tau(\mathbf{X})$.

Algorithm 1 Semi-parametric Double-BF-DNN Procedure

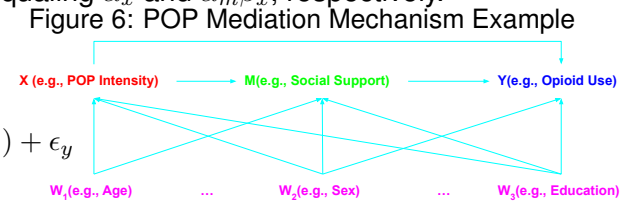
1: Approximate $e(\mathbf{X})$ by using a BF-DNN approximator $\widehat{e}(\mathbf{X})$.
2: Estimate β_1 via the PS covariate adjustment method.
3: Approximate $\xi^*(\mathbf{X}, \beta_1)$ based on a BF-DNN approximator $\widehat{\xi}^*(\mathbf{X})$.
4: Estimate β as $\widehat{\beta}^*$, and $V(\widehat{\beta}^*)$ as $\widehat{V}(\widehat{\beta}^*)$.

Aim 2b: BF-DNN semi-parametric mediation analysis Let Y be an outcome (e.g., opioid use), X be a covariate of primary interest (e.g., POP intensity), M be the mediator (e.g., social support), and $\mathbf{W}$ be a p-dimensional moderator (e.g., age, sex, education, etc) that may confound the relationships among the three model variables. We will employ the causal-model / potential-outcomes framework [123, 130–133] which provides a principled approach, with clearly stated assumptions on mediation in terms of conceived manipulations. In the potential outcome framework, the estimands of interest are defined by nested potential outcomes [130], via a causal model illustrated in Figure 6. Following Albert [134], we denote $\mathbf{W_1}$, $\mathbf{W_2}$, and $\mathbf{W_3}$ as the confounders of the $X \to M$, $X \to Y$, and $M \to Y$ relationships, respectively, and $\mathbf{W_{j,j'}}$ as the vector containing the union of covariates given by $\mathbf{W_j}$ and $\mathbf{W_{j'}}$. For continuous Y and M with the causal model (Figure 6), we extend the parametric mediation analysis models [134, 135] to the following semi-parametric regression models:

$$Y = \alpha_0 + \alpha_x X + \alpha_m M + \eta(\mathbf{W_{2,3}}) + \epsilon_y$$
$$M = \beta_0 + \beta_x X + \xi(\mathbf{W_{1,3}}) + \epsilon_m$$

where $\eta(.)$ and $\xi(.)$ are unknown smooth functions of $\mathbf{W}$, ϵ_y and ϵ_m are random errors associated with Y and M, respectively. The direct and indirect effect at reference exposure level x are defined as $D(x) = E\{Y(1, M(x))\} - E\{Y(0, M(x))\}$ and $I(x) = E\{Y(x, M(1))\} - E\{Y(x, M(0))\}$, equaling α_x and $\alpha_m \beta_x$, respectively.

Figure 6: POP Mediation Mechanism Example



To estimate α_x, α_m and β_x, we again consider the semi-parametric modeling strategy in Robinson [67] where the estimates are obtained from the following regression models

$$Y - E(\widehat{Y|\mathbf{W_{2,3}}}) = \alpha_x(X - E(\widehat{X|\mathbf{W_{2,3}}})) + \alpha_m(M - E(\widehat{M|\mathbf{W_{2,3}}})) + \epsilon_y$$
$$M - E(\widehat{M|\mathbf{W_{1,3}}}) = \beta_x(X - E(\widehat{X|\mathbf{W_{1,3}}})) + \epsilon_m.$$

where $E(\widehat{Y|\mathbf{W_{2,3}}})$, $E(\widehat{M|\mathbf{W_{2,3}}})$, $E(\widehat{X|\mathbf{W_{2,3}}})$, $E(\widehat{M|\mathbf{W_{1,3}}})$, and $E(\widehat{X|\mathbf{W_{1,3}}})$ are the semi-parametric estimators of $E(Y|\mathbf{W_{2,3}})$, $E(M|\mathbf{W_{2,3}})$, $E(X|\mathbf{W_{2,3}})$, $E(M|\mathbf{W_{1,3}})$, and $E(X|\mathbf{W_{1,3}})$ estimated by the BF-DNN, respectively. For generalized linear models on Y and M, we will similarly extend the approach in Albert [134] to estimate the (natural) direct and indirect effects semi-parametrically.

Preliminary studies of Aim 2 methods on real EHR data: To investigate the proposed Double-BF-DNN method for adjusting confounding factors, we applied it to a real POP EHR data set extracted from University of Florida's Integrated Data Repository, i.e., a large EHR database similar to CDW-H. It included 3196 patients underwent the digestive, musculoskeletal, or nervous system surgeries. We compared two anesthetic methods: nerve block ($Z = 1$) vs general anesthesia ($Z = 0$) for relieving the severest pain within 72 hours after surgery, which is quantified numerically and scaled between 0 and 10, where the higher pain scores corresponded to more severe pain. Among the 3196 patients, 2438 used the nerve block procedure while the rest opted for general